1 Profit = \$400 − \$320 = \$80 or $\frac{400}{320} \times 100$

[M1]

25 **[A1]**

[Total Marks: 2]

2 **(a)** $\frac{15}{100} \times 80$ or 12 **[M1]**

68 **[A1]**

(b) $\frac{4000 - 3200}{4000} \times 100$ or $\frac{800}{4000} \times 100$ **[M1]**

20 **[A1]**

[Total Marks: 4]

3 **(a)** $\frac{24}{60} \times 100$ **[M1]**

40 **[A1]**

(b) $\frac{15}{100} \times 80 = 12$ OR 80×0.85 **[M1]**

68 **[A1]**

[Total Marks: 4]

4 **(a)** $\frac{15}{100} \times 320$ or $10\% = 32, 5\% = 16$ **[M1]**

48 **[A1]**

(b) $\frac{20}{100} \times 45 = 9$ or $\frac{80}{100} \times 45$ **[M1]**

$45 - 9$ (if first method used) **[M1]**

36 **[A1]**

[Total Marks: 5]

5 **(a)** Discount $= 0.15 \times 120 = 18$ OR 0.85×120 **[M1]**

102 **[A1]**

(b) Let cost price be C. $1.2C = 120$ **[M1]**

$C = \frac{120}{1.2}$ **[M1]**

100 **[A1]**

[Total Marks: 5]

6 **(a)** $\frac{15}{100} \times 240$ or 0.15×240 **[M1]**

36 **[A1]**

(b) $\frac{40-32}{40} \times 100$ or $\frac{8}{40} \times 100$ or $100 - \frac{32}{40} \times 100$ **[M1]**

20 **[A1]**

(c) $120\% = 60$ or $1.2x = 60$ **[M1]**

50 **[A1]**

[Total Marks: 6]

7 **(a)** M1 for $15 \times (1 - 0.20)$ or $15 - 3$

12 **[A1]**

(b) M2 for $56 \div 0.8$ or $\frac{56}{80} \times 100$
(M1 for $80\% = 56$ or equivalent)

70 **[A1]**

(c) M1 for $\frac{42-30}{30} \times 100$ or $\frac{12}{30} \times 100$

40 **[A1]**

[Total Marks: 7]

8 **(a)** $4500 \times \left(1 + \frac{3.2}{100}\right)^5$ **[M2]**

OR 4500×1.032^5 **[M2]**

(M1 for 4500×1.032^k where $k > 1$)

5267.57... **[A1]**

(b) $\frac{408-320}{320} \times 100$ or $\frac{88}{320} \times 100$ **[M1]**

27.5 **[A1]**

[Total Marks: 5]

9 **(a)** $45\,000 \times 1.042$ OR $45\,000 + 45\,000 \times 0.042$ **[M1]**

46 890 **[A1]**

(b) $100\% - 24\% = 76\%$ **[B1]**

$\frac{342}{0.76}$ OR $\frac{342}{76} \times 100$ **[M1]**

450 **[A1]**

[Total Marks: 5]

10 **(a)** $4500 \times (1 + \frac{2.4}{100})^5$ or 4500×1.024^5 **[M2]**

If M0, M1 for 4500×1.024 or 4608 seen.

5066.57 or 5066.56... **[A1]**

(b) Interest earned $= 5066.57 - 4500 = 566.57$ (accept unrounded) **[M1]**

$\frac{4500 \times r \times 5}{100} = 566.57$ OR $225r = 566.57$ **[M1]**

2.52 or 2.518... **[A1]**

[Total Marks: 6]

11 **(a)** $8000 \times \left(1 + \frac{3.5}{100}\right)^4$ or 8000×1.035^4 **[M2]**

If M0 scored, allow [M1] for 8000×1.035^k where $k > 1$

9180.18 **[A1]**

(b) $x \times \left(1 - \frac{4}{100}\right)^3 = 45000$ or $x \times 0.96^3 = 45000$ **[M1]**

$\frac{45000}{0.96^3}$ **[M1]**

50863 **[A1]**

[Total Marks: 6]

12 **(a)** $4500 \times \left(1 + \frac{2.4}{100}\right)^5$ or 4500×1.024^5 **[M2]**

[M1] for 4500×1.024^n where $n > 1$

5066.55 **[A1]**

(b) $100\% - 18\% = 82\%$ or 0.82 **[B1]**

$\frac{557.60}{0.82}$ or $x \times 0.82 = 557.60$ **[M1]**

680 **[A1]**

[Total Marks: 6]

13 **(a)** M2 for $8000 \times \left(1 + \frac{3.5}{100}\right)^4$

(M1 for 8000×1.035^k where $k > 1$)

9180.19 **[A1]**

(b) M2 for $24000 \times 0.88^n < 10000$ or evaluating at least two consecutive trials (e.g. $n = 6$ and $n = 7$)

(M1 for 24000×0.88^n seen)

A1 for $n = 7$ (Value at year 6 is \$11145.69, Value at year 7 is \$9808.21)

(c) M2 for $150000 \div 1.25$

(M1 for 125% = 150000 or equivalent)

120000 **[A1]**

[Total Marks: 9]

14 **(a)** $4000 \times \left(1 + \dfrac{2.5}{100}\right)^5 - 4000$ [M2]

(M1 for 4000×1.025^5)

525.63 or 525.64 **[A1]**

(b) Interest $= 4896 - 4000 = 896$ **[B1]**

$896 = 4000 \times \dfrac{3.2}{100} \times t$ or $t = \dfrac{896}{128}$ **[M1]**

7 **[A1]**

(c) $P \times 1.018^6 = 6676.70$ **[M1]**

$P = \dfrac{6676.70}{1.018^6}$ **[M1]**

6000 **[A1]**

[Total Marks: 9]

15 **(a)** $(1.25 \times 10^6) \times 1.045^{10}$ **[M1]**

1941214.7... **[A1]**

1.94×10^6 **[A1]**

(b) $299.25 = 86\%$ or $\dfrac{299.25}{1 - 0.14}$ **[M1]**

348 **[A1]**

(c) Let V be the original value. $V \times 0.85 \times 0.90^2 = 17212.50$ **[M1]**

$V = \dfrac{17212.50}{0.85 \times 0.81} = 25000$ **[A1]**

[Total Marks: 7]

16 **(a)** $1\,250\,000 \times (1 + \dfrac{x}{100})^3 = 1\,367\,631$ **[M1]**

$(1 + \dfrac{x}{100})^3 = 1.0941048$ **[M1]**

$1 + \dfrac{x}{100} = \sqrt[3]{1.0941048} = 1.0304\cdots \implies x = 3.04\cdots \approx 3$ **[A1]**

(b) Target population $= 2 \times 1\,250\,000 = 2\,500\,000$ **[B1]**

$1\,250\,000 \times (1.03)^n > 2\,500\,000 \implies 1.03^n > 2$ **[M1]**

Using logs: $n \log(1.03) > \log(2)$ OR trial and improvement yielding n between 23 and 24 **[M1]**

$n > 23.449\ldots$, so it takes 24 years. Year $= 2015 + 24 = 2039$ **[A1]**

[Total Marks: 7]

17 **(a)** $p \times 1.2 = 960$ or $\dfrac{960}{1.2}$ **[M1]**

800 **[A1]**

(b) Selling price $= 1.15q$ **[B1]**

Marked Price $\times 0.9 = 1.15q$ or $\dfrac{1.15q}{0.9}$ **[M1]**

$\dfrac{23q}{18}$ or $\dfrac{23}{18}q$ **[A1]**

(c) $500 \times \left(1 + \dfrac{r}{100}\right)^{10} = 740.12$ or $\left(1 + \dfrac{r}{100}\right)^{10} = \dfrac{740.12}{500}$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[10]{1.48024}$ or 1.04 **[M1]**

4 **[A1]**

[Total Marks: 8]

18 **(a)** $5000 \times \left(1 + \dfrac{2.5}{100}\right)^3$ **[M1]**

5384.45 **[A1]**

(b) $\left(1 + \dfrac{r}{100}\right)^5 = 1.30$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[5]{1.30}$ **[M1]**

5.39 (or $5.387\ldots$) **[A1]**

(c) $V \times (1 - 0.12)^3 = 17\,036.80$ **[M1]**

$V = \dfrac{17\,036.80}{0.88^3}$ **[M1]**

25 000 **[A1]**

[Total Marks: 8]

19 **(a)** Expression for 3 years depreciation: e.g. $V \times (1 - 0.15) \times (1 - 0.08)^2$ or $V \times 0.85 \times 0.92^2$ **[M2]**

[M1] for $V \times 0.85$ or $V \times 0.92^2$ seen

$V \times 0.71944 = 17986$ or $V = \dfrac{17986}{0.85 \times 0.92^2}$ **[M1]**

25 000 **[A1]**

(b) $\dfrac{25000 - 17986}{25000} \times 100$ or $100\% - (0.85 \times 0.92^2 \times 100)\%$ **[M1]**

28.1 **[A1]**

(c) $P \times \left(1 + \dfrac{r}{100}\right)^{12} = 2P$ or $\left(1 + \dfrac{r}{100}\right)^{12} = 2$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[12]{2}$ or 1.059... **[M1]**

5.95 (correct to 3 significant figures) **[A1]**

[Total Marks: 9]

20 **(a)** $V \times (1 - 0.15)^3 = 12282.50$ or $V \times 0.85^3 = 12282.50$ [M2]
(M1 for $V \times 0.85^k = 12282.50$ where $k > 1$ or $\dfrac{12282.50}{0.85}$ seen)

20 000 **[A1]**

(b) $15000 \times \left(1 + \dfrac{x}{100}\right)^5 = 21500$ or $\left(1 + \dfrac{x}{100}\right)^5 = \dfrac{215}{150}$ **[M1]**

$\sqrt[5]{\dfrac{215}{150}} - 1$ or $1.0746... - 1$ **[M1]**

7.5 **[A1]**

(c) Multiplier $= 1.04 \times 0.98 \times 1.05$ or 1.07016 [M2]
(M1 for two correct multipliers used, e.g. 1.04×0.98)

Overall percentage increase $= (1.07016 - 1) \times 100$ **[M1]**

7.02 **[A1]**

[Total Marks: 10]

21 **(a)** M2 for $\left(1 + \dfrac{r}{100}\right)^6 = 2$ or equivalent
(M1 for $\left(1 + \dfrac{r}{100}\right)^6$ seen)

12.2 **[A1]**

(b) M2 for $X \times 1.40 \times 0.85 = 142.80$ or $1.19X = 142.80$
(M1 for $X \times 1.40$ or Selling Price $\times 0.85 = 142.80$)

120 **[A1]**

(c) M1 for $P \times \dfrac{x}{100} \times 5 = 250$ (implies $Px = 5000$)

M2 for $P\left(1 + \dfrac{x}{100}\right)^2 - P = 102.50$
(M1 for $P\left(1 + \dfrac{x}{100}\right)^2 = P + 102.50$)

M1 for correctly substituting to form an equation in one variable, e.g.
$\dfrac{5000}{x}\left(\dfrac{2x}{100} + \dfrac{x^2}{10000}\right) = 102.50$

$x = 5$ **[A1]**

$P = 1000$ **[A1]**

[Total Marks: 11]